Q1:JUN 2008 P2

7 **Fig. 7.1** shows a bridge rectifier circuit.

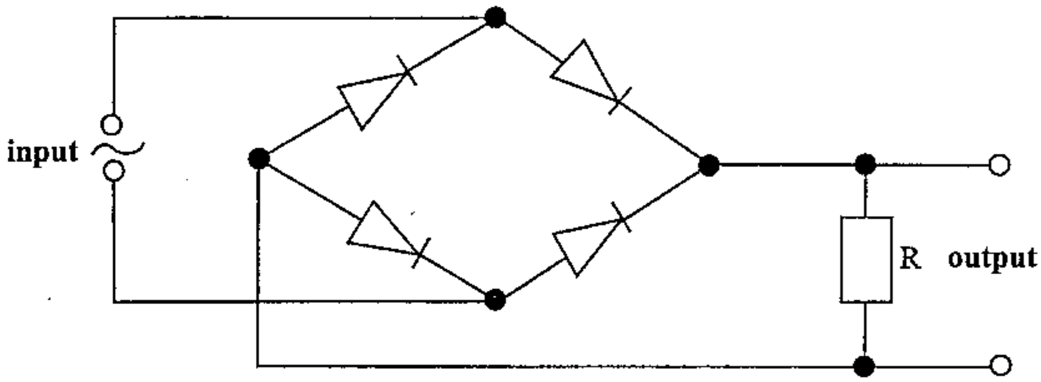

Fig. 7.1

(a) **(i)** On the same axes sketch graphs of input and output potential difference against time for two complete cycles. [2]

(ii) On **Fig. 7.1** add a reservoir capacitor to complete the smoothing circuit. [1]

(iii) On the sketch in **(a)(i)** include the output potential difference from smoothing circuit. [1]

(b) Explain the effect to the output potential and rectifier of using a capacitor of very large capacitance.

____

____

____

____ [1]

Q2:NOV 2014 P2

7 **(a)** Define *root mean square value of an alternating current.*

____

____ [1]

(b) **Fig. 7.1** shows a voltage wave on a cathode ray oscilloscope. The time-base and the y-gain are set at 5 ms/cm and 0.2 V/cm respectively.

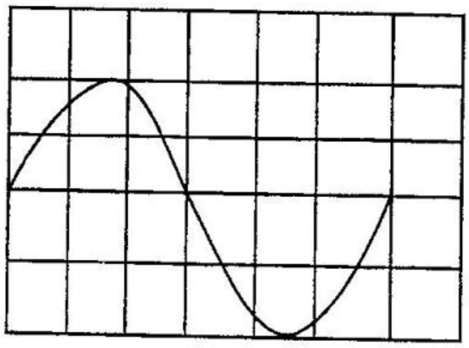

Fig. 7.1

Determine

(i) the peak-to-peak voltage,

____

(ii) the root mean square voltage,

____

(iii) the mean power developed in a 30 Ω resistor.

____

[5]

Q3:NOV 2015 QP 2

6 (a) Describe the principle of operation of an ideal transformer.

____

____

____ [3]

(b) The primary coil of an ideal transformer has 480 turns and is connected to a 120 V (r.m.s), 60 Hz sinusoidal supply. The secondary coil has 40 turns and is connected to a resistor of resistance 4.0 Ω as shown in **Fig. 6.1.**

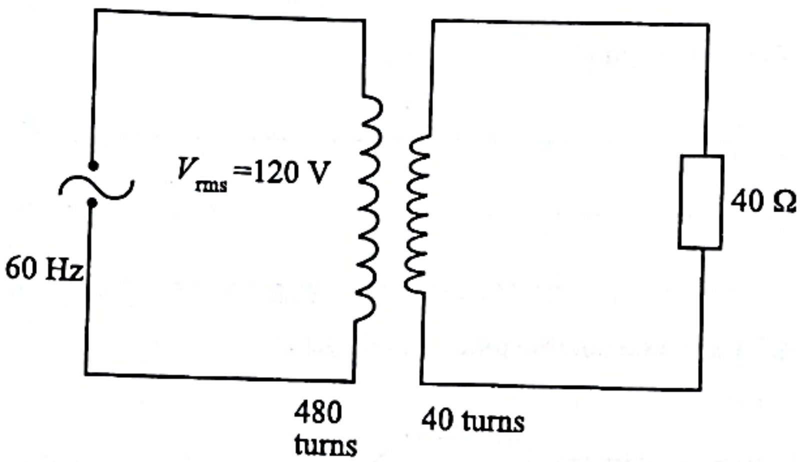

Fig.6.1.

Calculate

(i) V_{rms} across the resistor,

V_{rms} = ______________________

(ii) peak power dissipated by the resistor.

peak power ______________________________ [4]

Q4:JUN 2019 P2

3 (a) Explain how a transformer produces an output voltage greater than the input voltage.

____

____

____

____

____

______________________________ [3]

(b) **Fig. 3.1** shows a generator providing power to equipment rated at 130 kW, 240 V a.c. The generator is connected to the equipment by two conductors which have a total resistance of 0.30 Ω.

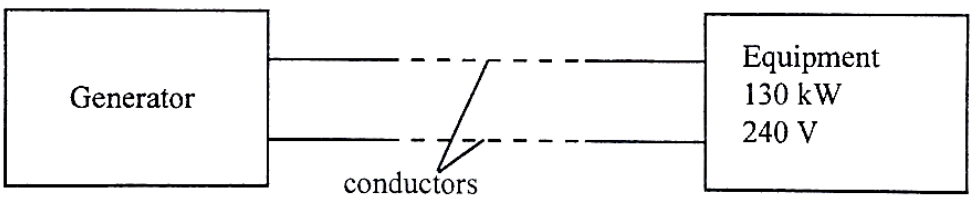

Fig. 3.1

The equipment is operating at its rated power.

Calculate the

(i) power loss in the conductors,

(ii) voltage developed by the generator,

voltage ____________

(iii) efficiency of the transmission system.

efficiency ____________ [5]

(c) Explain why transformers produce a buzzing sound when they are operating.

____

____

____

____

______________________________ [2]

Q5:JUN 2007 P3

4 (a) Define

(i) peak value,

(ii) root-mean-square value,

as applied to alternating voltage. [3]

(b) (i) Explain why the root-mean-square and not the peak or the mean value is preferred in describing alternating voltage.

(ii) State the relationship between the root-mean-square and the peak values for a sinusoidal voltage.

(iii) Deduce an equation for the average power for the voltage in **b(ii)** above. [7]

(c) A sinusoidal alternating voltage of 10 $V_{r.m.s.}$ and frequency 50 Hz is applied to a resistor of 5.0 Ω.

(i) Determine the root-mean-square current.

(ii) Calculate the maximum voltage. [2]

(d) The alternating voltage is now passed through a set of 4 diodes arranged to produce full wave rectification.

(i) Draw a circuit diagram showing the arrangement of the diodes and the flow of current in the resistor.

(ii) Sketch graphs to show the input and the output voltages of the circuit in **d(i)**.

(iii) A capacitor is used to smooth the rectified potential difference.

1. Add the capacitor to your diagram in **d(i)**.

2. State **one** property of a capacitor that qualifies it for the smoothing purpose.

3. Explain the operation of the capacitor in smoothing. [8]

Q6:JUN 2007 P3

7 (c) (i) Describe the structure of a step down transformer.

(ii) Explain the use of the iron core in a transformer. [4]

(d) A transformer is to be used as a 12 V DC battery charger. The transformer has a primary to secondary turns ratio of 16: 1. The input voltage to the transformer is 240 V.

(i) Calculate the output voltage.

(ii) Battery chargers use direct current. Suggest **modifications** that can be done to the transformer circuit so that it can charge a discharged accumulator.

(iii) The initial output current of the transformer when the battery starts to charge is 2.2 A. Calculate the current that is flowing in the primary coil of the transformer. [7]

(e) Explain briefly why oil is used in large power transformers. [2]

Q7:JUN 2009 P5

2 (c) (i) Give **one** condition that will enable a transformer to behave ideally.

(ii) Alternating voltage of peak value 10.5 V is the input of an ideal transformer. The mean power output from the transformer is supplied to a consumer at 0.440 kW. The peak current in the secondary coil is 10.0 A and the number of turns in the primary coil is 260.

Determine the number of turns in the secondary coil. [4]

Q8:NOV 2010 P5

1 (a) (i) Explain

1. peak value,
2. r.m.s value of an alternating voltage.

(ii) An alternating voltage is represented by $V = 200\cos 20\pi t$

Determine its

1. frequency,
2. peak value,
3. r.m.s value. [8]

(b) (i) Describe the principles of operation of an ideal transformer.

(ii) Electrical energy is transmitted as high alternating voltage.

Explain the advantages of using

1. alternating voltage,
2. high voltage. [8]

Q9:NOV 2012 P5

3 Fig. 3.1 shows a transformer circuit.

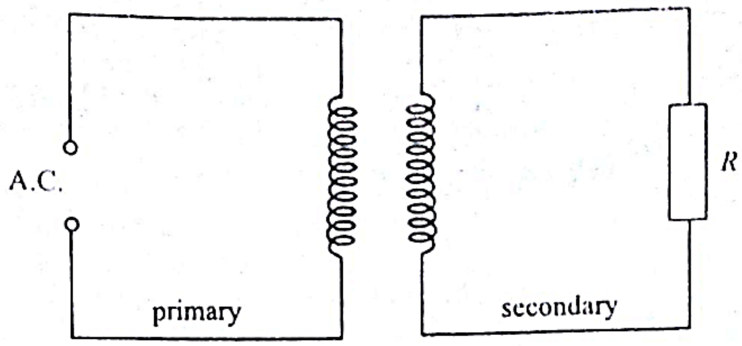

Fig 3.1

(a) Explain the effect on the current flowing in the primary circuit of

(i) a fall in the supply frequency,

(ii) a reduction in the number of turns in the secondary circuit.

(b) (i) Calculate the current which flows in a resistance of 40 Ω connected to a secondary coil of 60 turns if the primary has 1 200 turns and is connected to a 240 V.

(ii) State any **one** assumption made in making your calculation.

[7]

(c) The *Dragline* is a large current-drawing machine used in mining industries. Direct current motors have more torque to drive the dragline than alternating current motors. Alternating current from the mains is converted to direct current.

Describe and explain how direct current is maintained through the motor using four diodes. [5]

Q10:JUN 2013 P5

1 (d) (i) State and explain **two** features of a transformer which help minimise energy losses.

(ii) **Fig 1.2** shows the power variation with time for the primary coil of an ideal transformer.

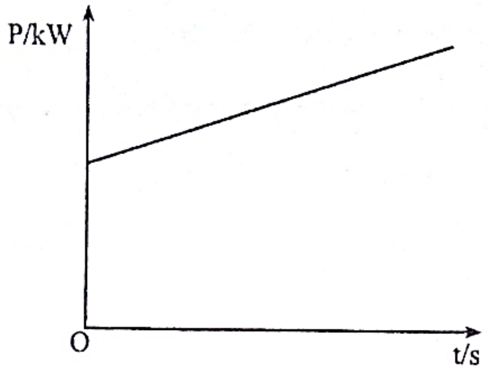

Fig. 1.2

Copy **Fig.1.2** and sketch the power variation for the secondary coil on the same axes.

[5]

Q11:NOV 2015 P5

2 **(b)** **Fig.2.1** illustrates a transformer.

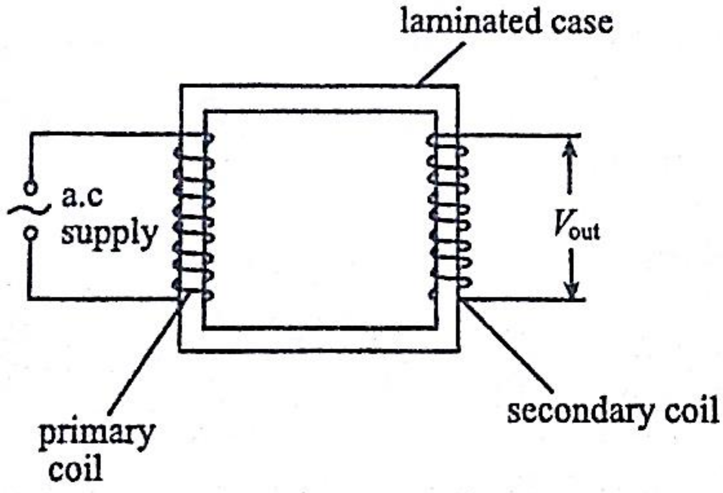

Fig.2.1

(i) Explain why

1. a soft iron core is used,

2. the core is laminated,

3. V_{out} and e.m.f. of the supply are out of phase,

4. thermal energy is generated in the core.

(ii) Give reasons why electrical energy is transmitted as

1. high voltage,

2. alternating current.

[8]

(c) Explain, using Lenz's law, why it is easier to switch on a bicycle dynamo when the lamps are switched off rather than when they are on. [2]

Q12:NOV 2011

1 (b) (i) **Fig. 1.2** shows the output of an alternating current source.

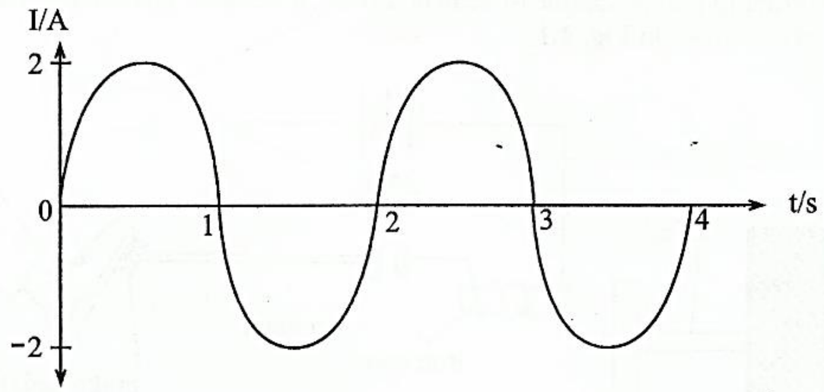

Fig. 1.2

1. State what is meant by *root mean square value* of alternating current.

2. Determine the average value of the current over the two cycles shown in **Fig. 1.2**.

(ii) When an alternating current source is connected to a hospital lamp, in an intensive care unit, the lamp lights with the same brightness as it does when connected across a 24 V battery.

Find the

1. root mean square voltage of the alternating current supply.

2. peak voltage of the alternating current supply.

(iii) Zimbabwe Electricity Supply Authority (ZESA) supplies electrical energy to a distant farm through power lines which have a total resistance of 5 Ω.

1. Explain why an input potential difference of 100 kV r.m.s. would be preferred to that of 10 kV r.m.s.

2. Calculate, in each case, the output voltage when the power input is 20 MW.

[12]

Q13:JUN 2017 P5

1 (a) (i) Distinguish between the rms value and peak value of an alternating voltage.

(ii) The voltage of the domestic mains supply is quoted as 240V.

1. State whether the voltage is the rms or peak value.

2. Calculate the maximum current which flows through a lamp rated 60W connected across this supply.

(iii) Give **one** economic advantage of using high voltage for power transmission on the National Grid.

[6]

Q14:JUN 2017 P5

3 (b) An ideal transformer is to be used to step down a 220 V a.c. supply to 7.5 V.

(i) Calculate the number of turns on the primary coil given that there are 140 turns on the secondary coil.

(ii) Suggest why the secondary coil wire should be thicker than the primary coil wire.

[3]

(c) State any **two** causes of power loss in a real transformer and explain how any **one** of them can be minimised. [3]

Q15:NOV 2018 P3

3 (d) (i) State any **two** reasons why oil is used in a transformer.

(ii) State and explain the economic advantage of using high voltages in the transmission of electricity.

(d) (iii) The value of alternating current generated by a source is given by

$$I = 15 \sin 6t.$$

The current flows through a 20 Ω resistor.

1. State the peak current value.

2. Calculate the period of the alternating current.

3. Determine the mean power dissipated by the 20 Ω resistor. [8]

Q16:2018 SPEC P3

3 (c) (i) State the function of a transformer.

(ii) A transformer has 600 turns in its primary coil and a sinusoidal input voltage of r.m.s. value 230 V. The output has a peak to peak voltage of 120 V.

1. Calculate the number of turns in the secondary coil.

2. Explain why r.m.s. values are considered and not mean values in a.c. circuits.

(iii) In a transformer, there is thermal energy produced. Describe and explain the sources of the thermal energy.

[9]